



天津医科大学
TIANJIN MEDICAL UNIVERSITY

脑功能影像后处理培训班

脑功能成像分析方法

于春水

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TIANJIN MEDICAL UNIVERSITY

任务态fMRI分析方法



Seiji Ogawa

Proc. Natl. Acad. Sci. USA
Vol. 87, pp. 9868–9872, December 1990
Biophysics

Brain magnetic resonance imaging with contrast dependent on blood oxygenation

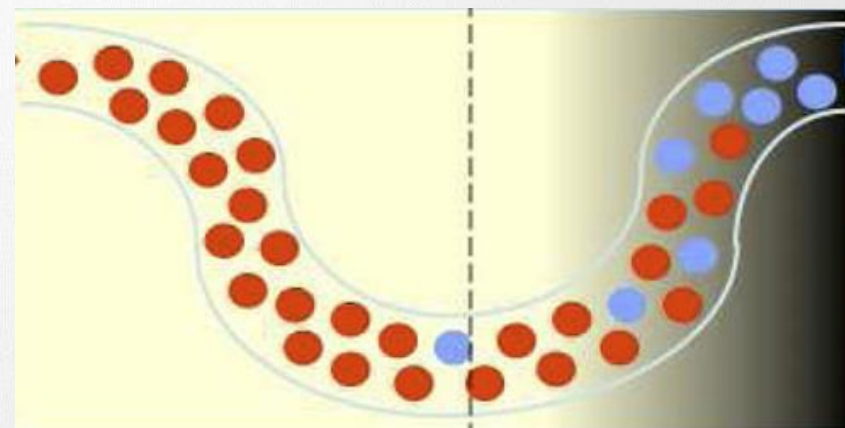
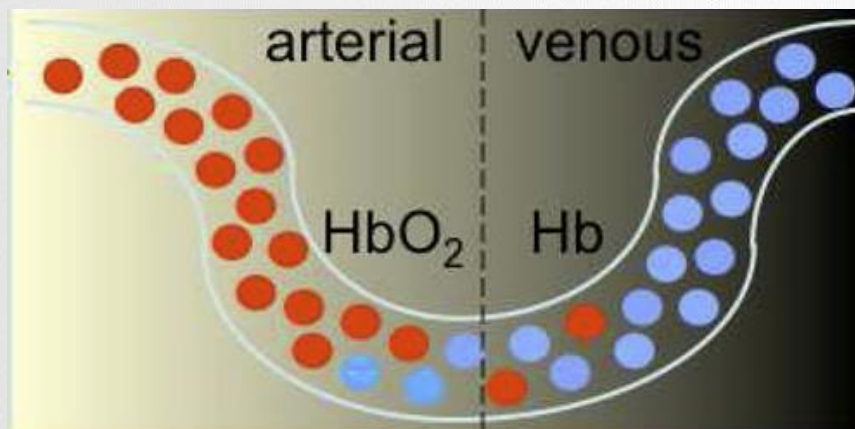
(cerebral blood flow/brain metabolism/oxygenation)

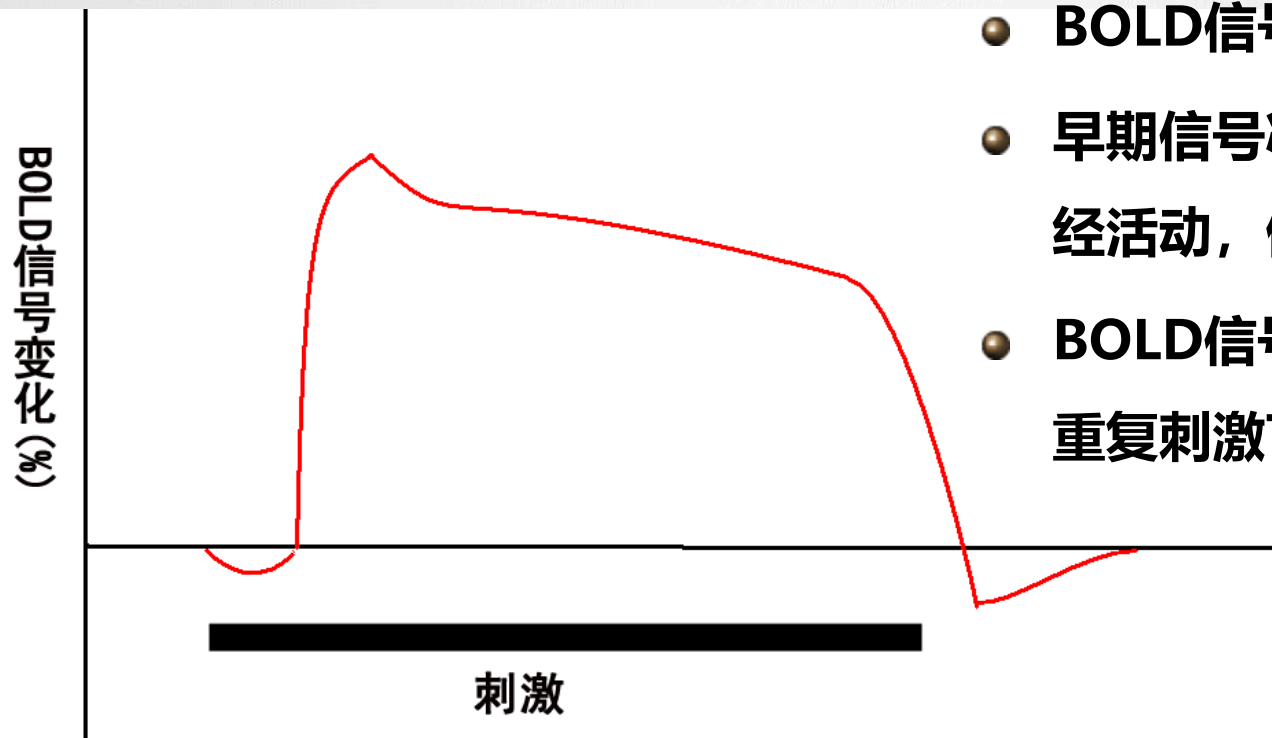
S. OGAWA, T. M. LEE, A. R. KAY, AND D. W. TANK

Biophysics Research Department, AT&T Bell Laboratories, Murray Hill, NJ 07974

Communicated by Frank H. Stillinger, September 24, 1990 (received for review August 1, 1990)

- 神经元活动（动作电位）消耗能量（糖和氧）
- 通过神经递质介导的信号通路使血管扩张和血流增加
- 神经血管耦合单元：神经元、星形细胞和内皮细胞
- 局部氧供给大于消耗，血管内脱氧血红蛋白净减少
- 脱氧血红蛋白的顺磁性导致局部磁场不均匀-T2*下降
- 脱氧血红蛋白减少造成信号增高





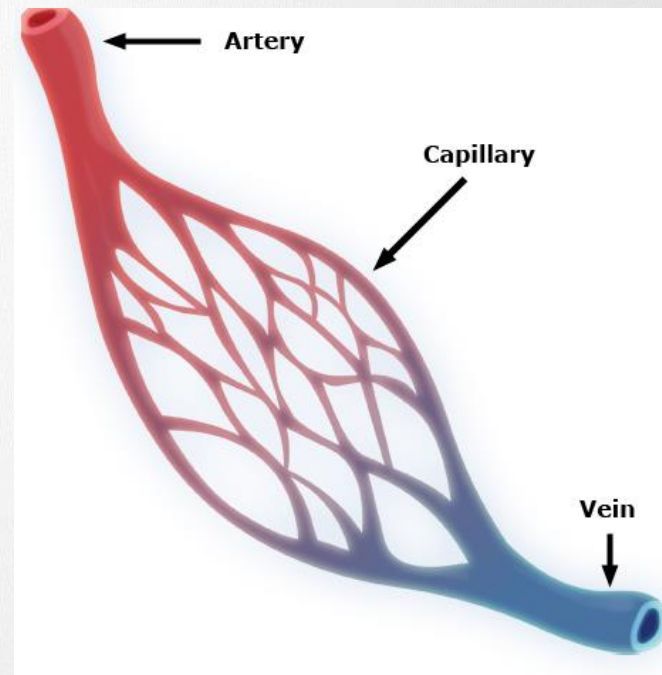
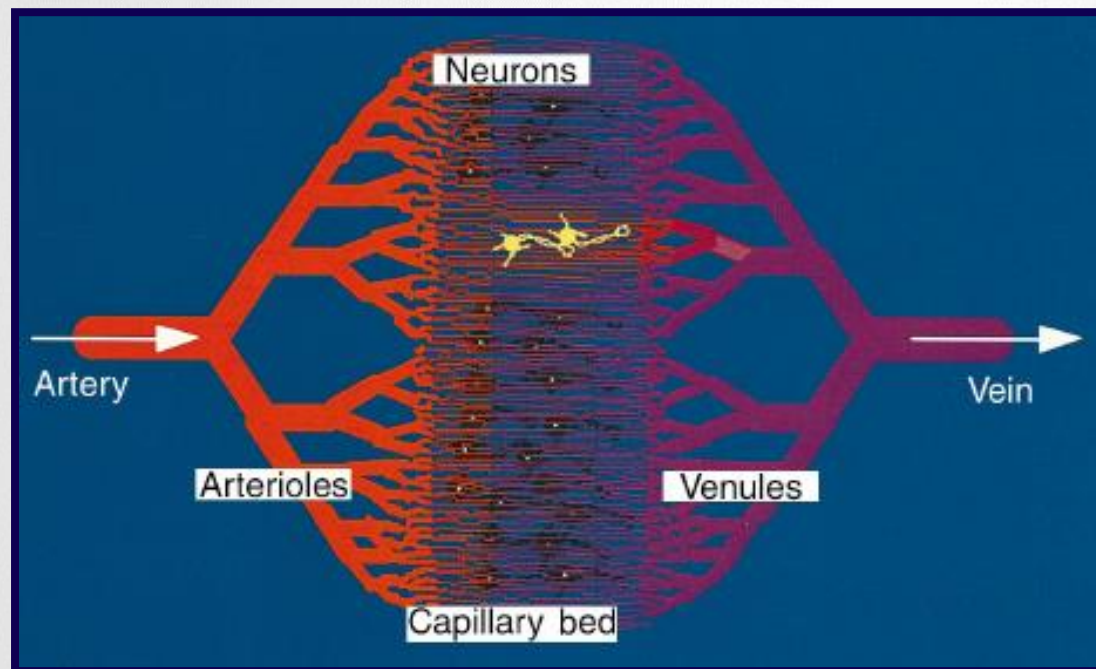
- BOLD信号变化微弱, $<5\%$
- 早期信号减低能够更好反应神经活动, 但太弱
- BOLD信号受噪声影响, 多次重复刺激可提高SNR

- 血液动力学反应持续10秒, 4-6秒达到高峰
- 血液动力学反应落后于神经元活动1-2秒
- 连续刺激导致神经元持续放电, 峰值达到一个平台
- 刺激停止后出现一个低射状态

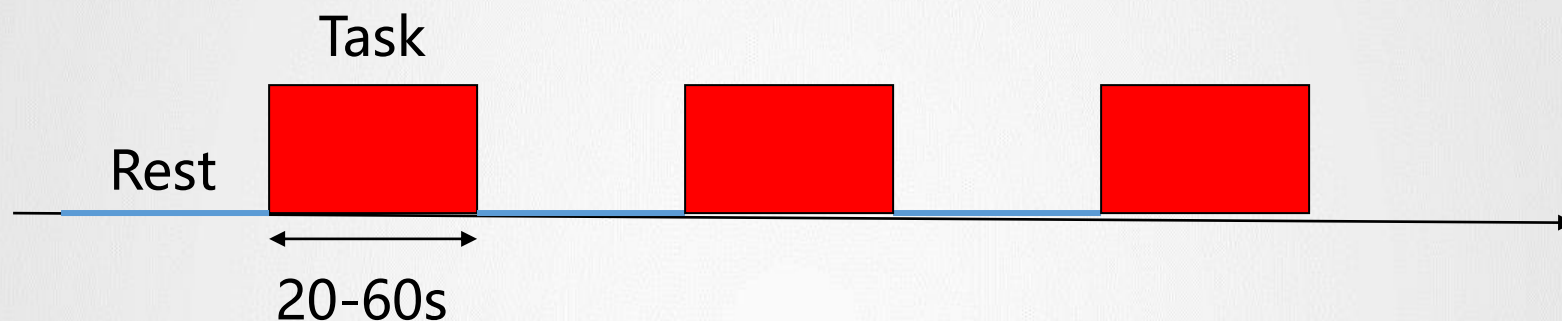
BOLD信号空间分辨率



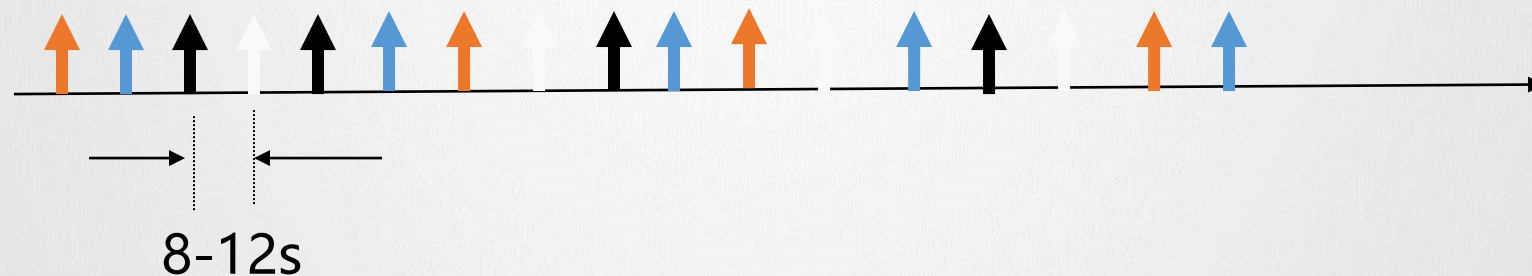
- BOLD信号变化可出现在神经元附近毛细血管网 (2%)
- 也可出现在较远的引流静脉 (10%)
- 提高空间分辨率可以抑制引流静脉成分
- 空间分辨率增高会降低SNR



组块Block-Design fMRI



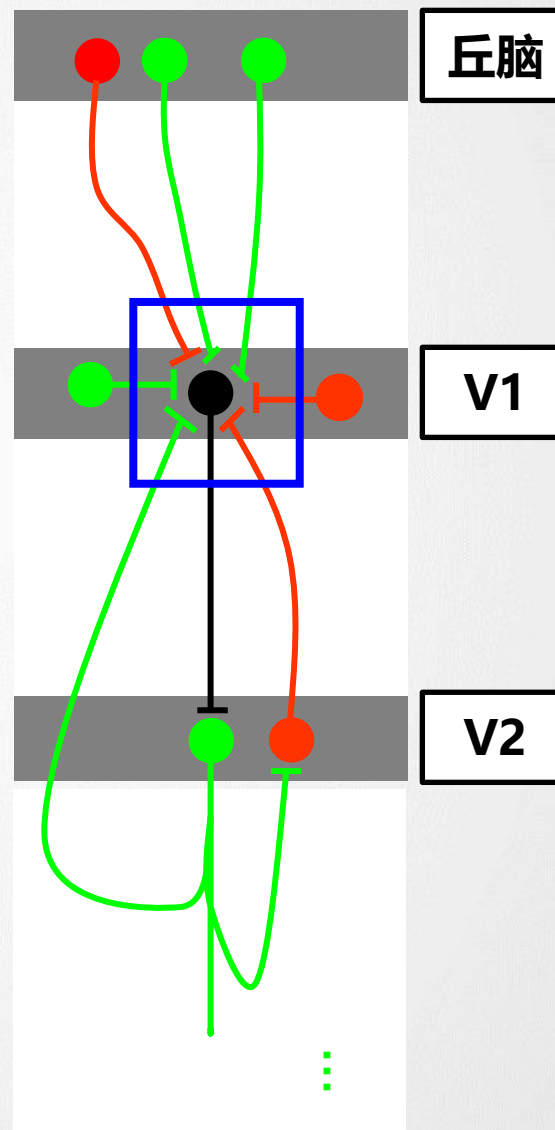
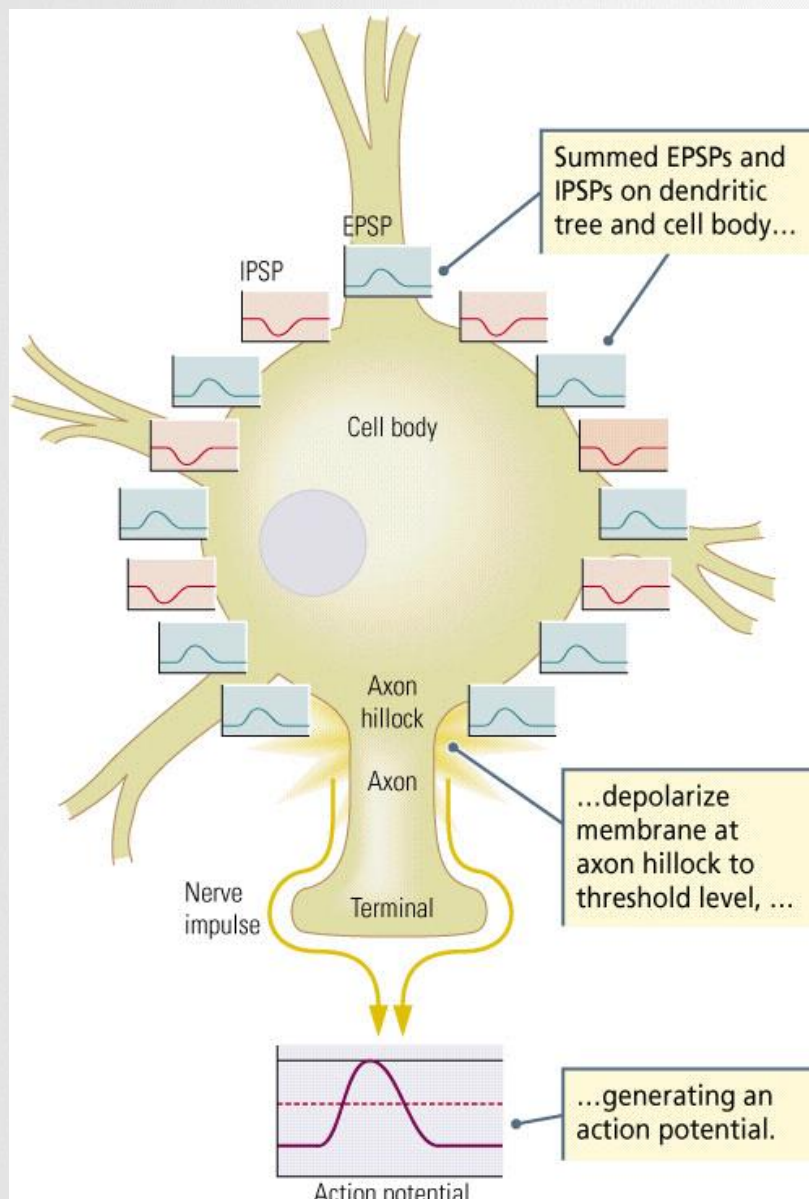
事件相关Event-Related fMRI



神经元活动的复杂性



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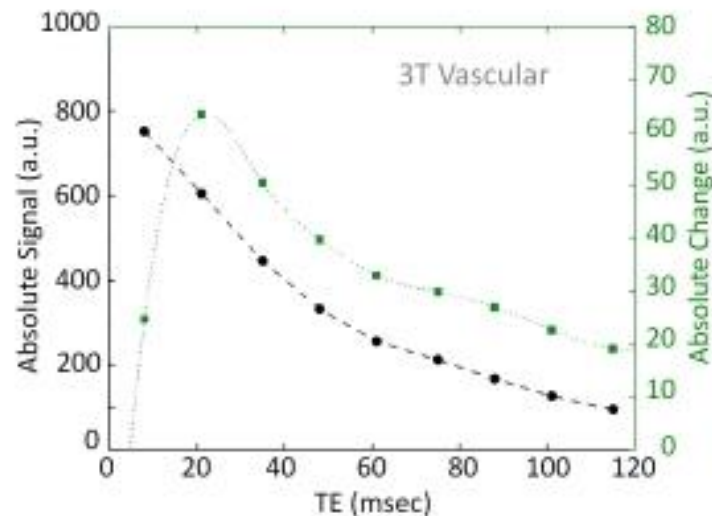
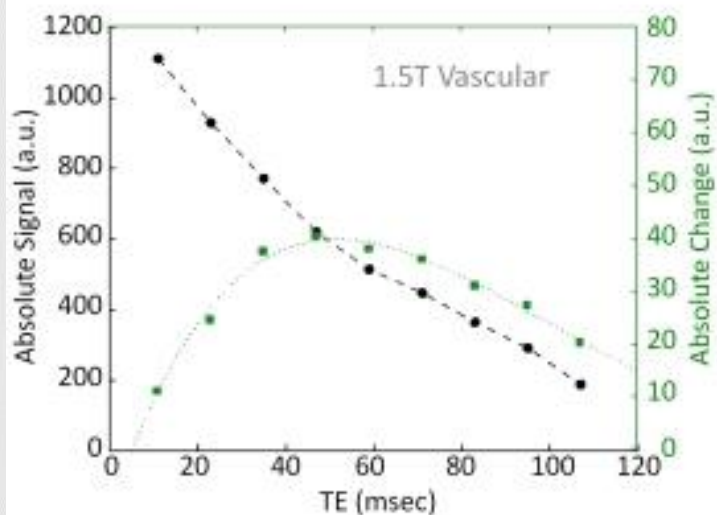
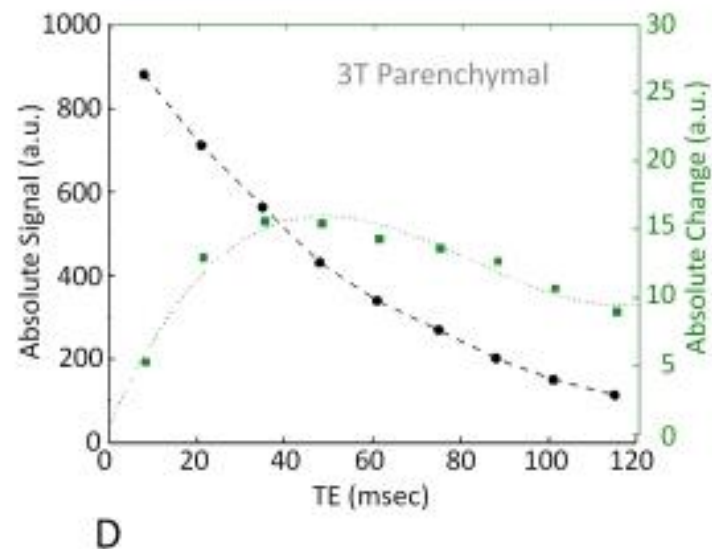
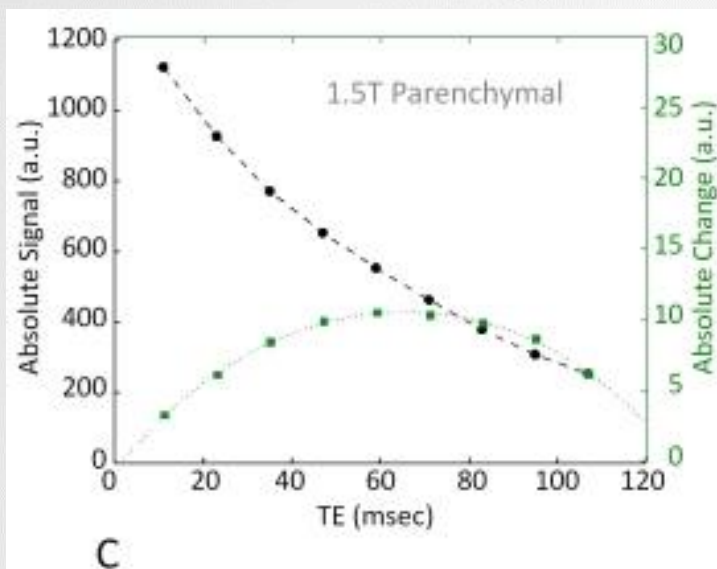


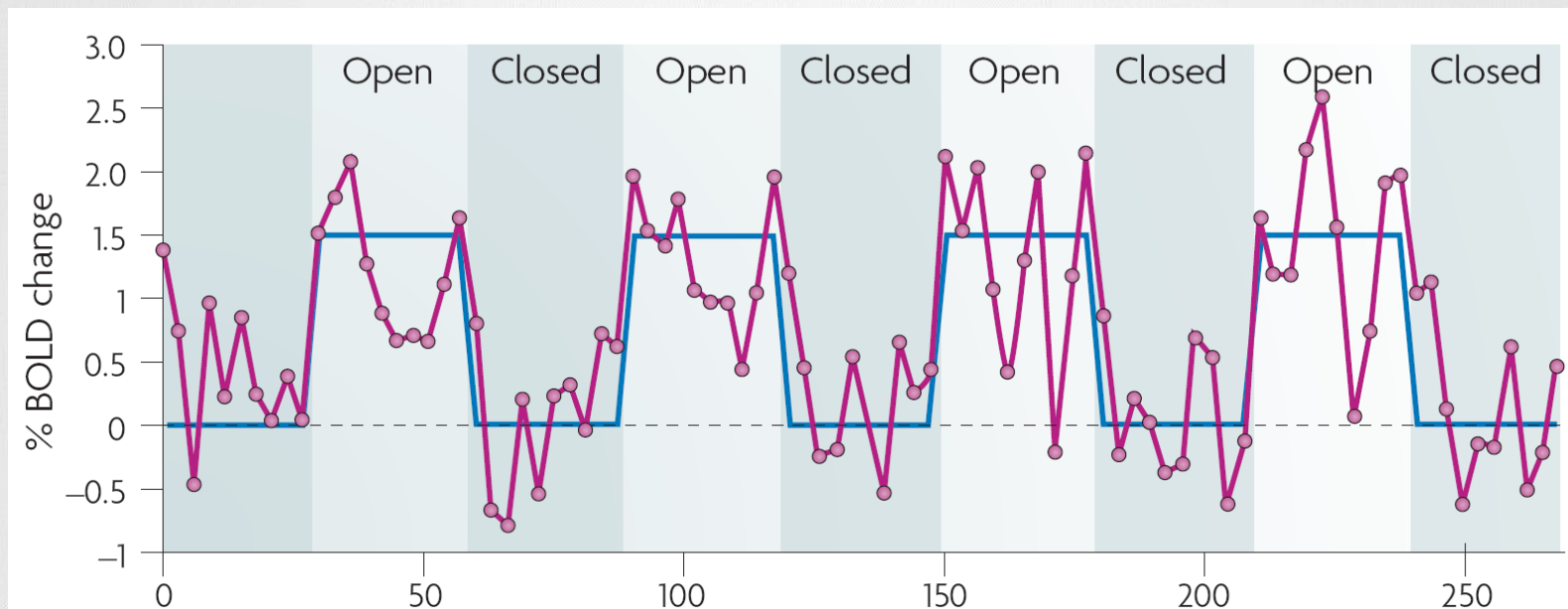
- **BOLD信号与局部场电位相关：与传入和中间神经元活动关系强，与传出神经元活动关系较弱**
- **不能区分前馈和反馈、兴奋和抑制性活动**
- **BOLD信号反应群体神经元活动**
- **信号强度取决于活动神经元的数量和幅度**
- **大量神经元轻微活动 > 少数神经元明显活动**
- **BOLD信号与单个神经元活动不匹配**

- 神经元活动增强--BOLD信号增高
- 复杂认知任务早期：BOLD信号与任务表现正相关
- 任务熟悉后：BOLD信号与表现负相关---效率假说
- 脑区之间神经元密度和血液动力学特性不同—不能比较
- 光遗传技术可精确控制神经元放电的同时检测BOLD变化
- 证明神经元放电与BOLD信号相关---线性迭加假设

- **热噪声：温度、线圈接收频率及电阻----影响所有体素**
- **系统噪声：设备硬件，部分可调节**
- **生理噪声：**
 - 呼吸、心跳、脑脊液搏动
 - 头动、紧张、烦躁
 - 血流、血氧动态变化
- **随机神经元活动**
- **反应策略的个体差异**

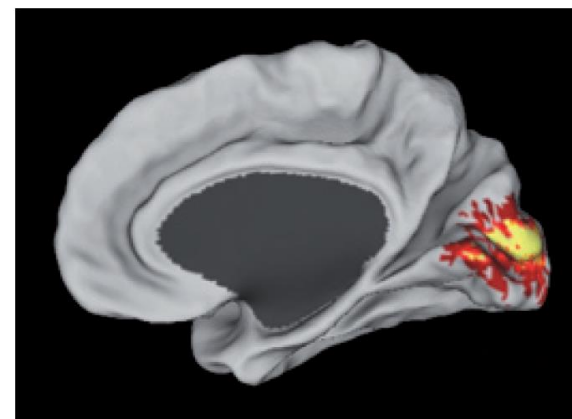
BOLD-fMRI优化





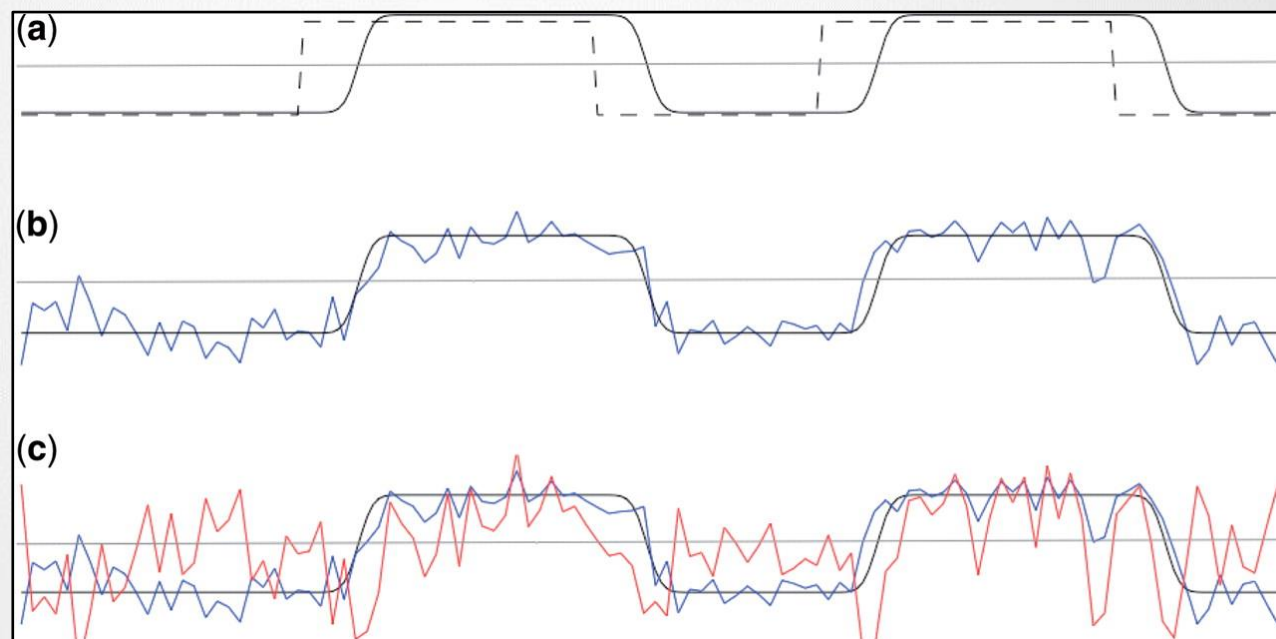
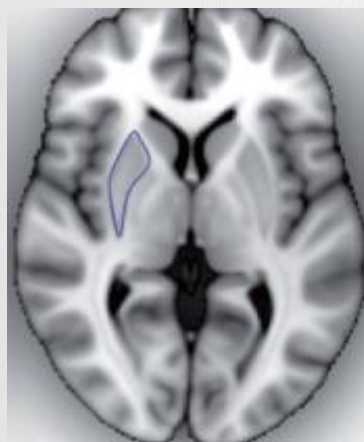
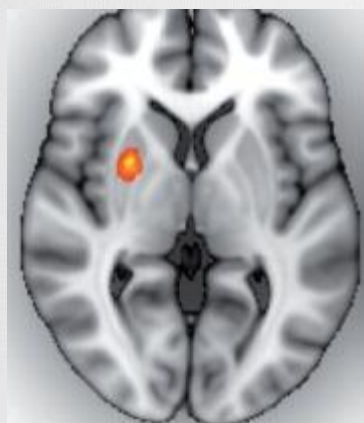
激活 = 刺激 - 对照

Open - Closed =



- 研究两个远隔脑区活动的相关性是否因心理状态不同而不同，既心理状态与功能连接有无交互作用

$$\text{PPI} = \text{PSY} \times \text{PHYS}$$

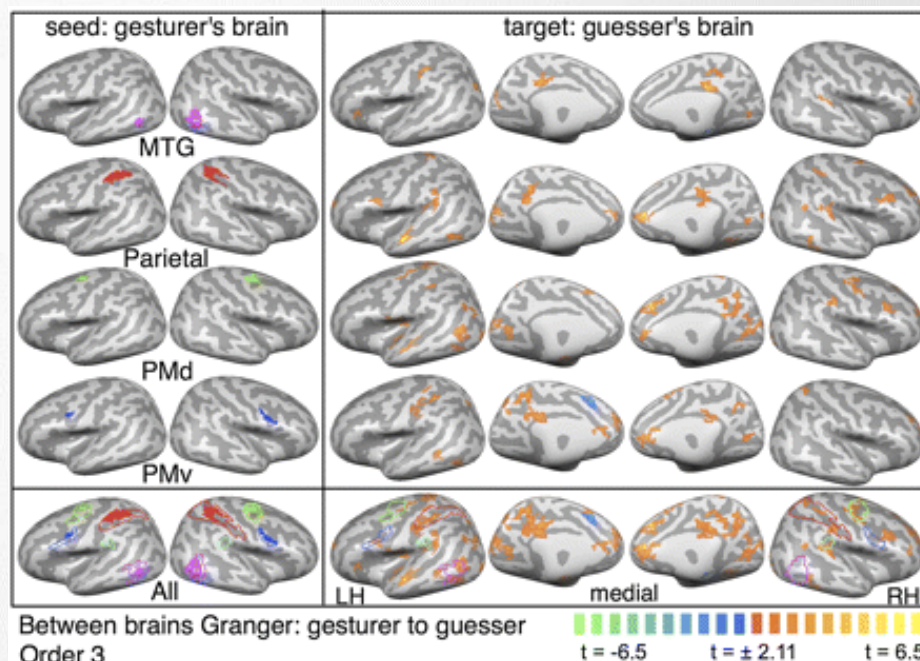
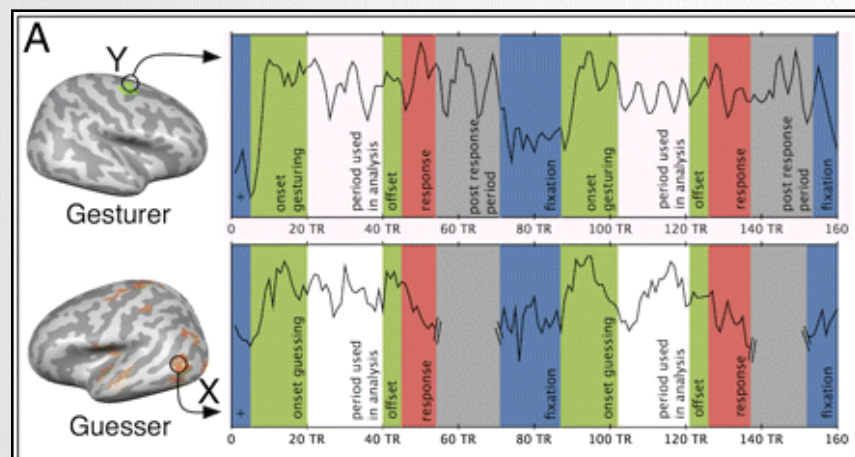


- **Granger causality mapping (GCM)**
Roebroeck et al., Neuroimage 2005
- **Structural equation modeling (SEM)**
- **Dynamic causal modeling (DCM)**
Friston et al., Neuroimage 2003

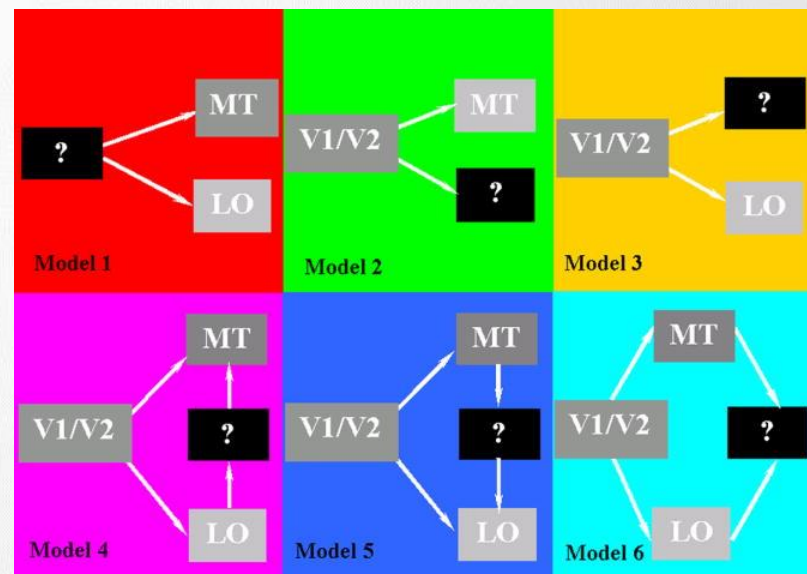
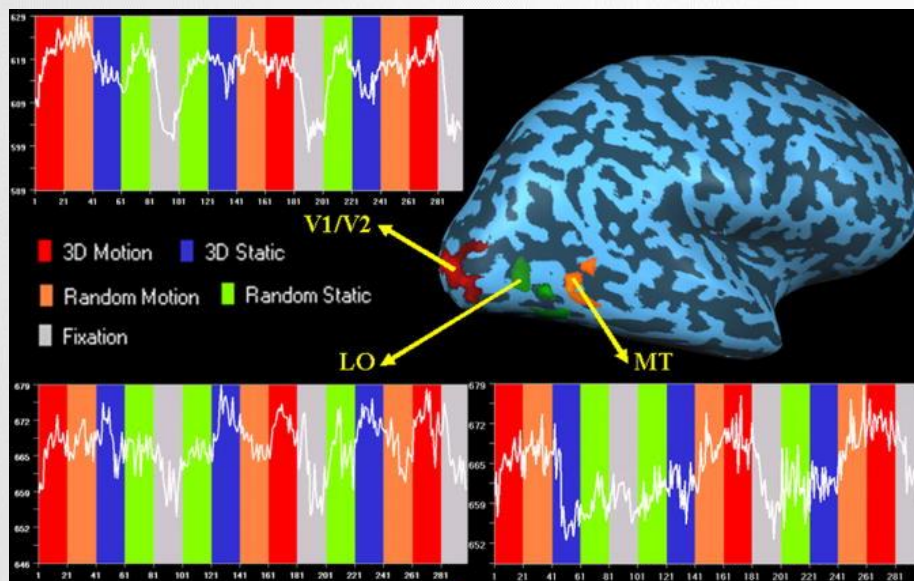
Granger causality mapping



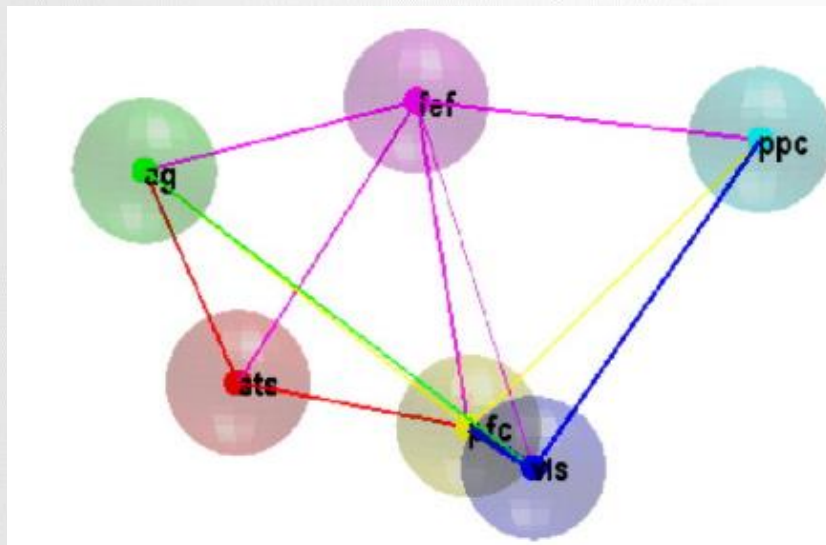
- 线性多变量自回归模型进行定量预测
- 无需先验解剖假设
- 用于探索性分析
- 受血液动力学特性的影响



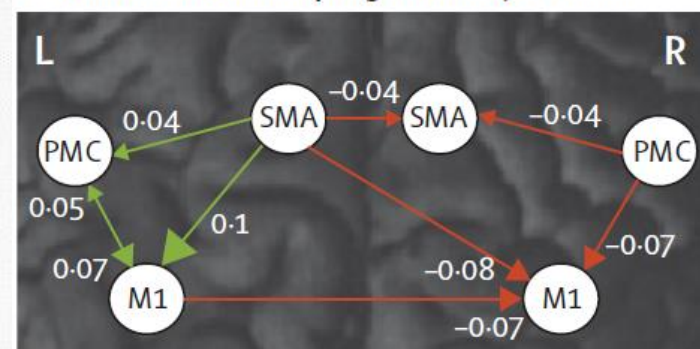
- 基于**假设**选择ROI
- 基于假设选择关系图
- 探索ROI之间的相互作用
- 不足：忽略重要脑区时会导致错误结果



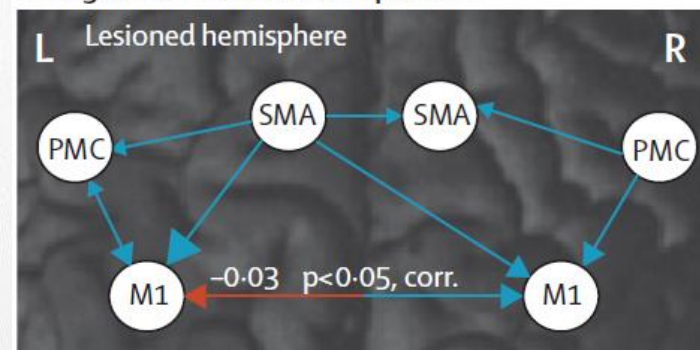
- 假设人脑是一个非线性动态系统
- 输入是已知的而不是随机的
- 研究输入对脑活动及脑连接的影响
- 验证具体**假设**，不适合探索性分析



A Modulations of coupling in healthy controls



B Significant difference in patients





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静息态fMRI分析方法

BOLD信号的自发性波动，在任务fMRI实验中当作噪声

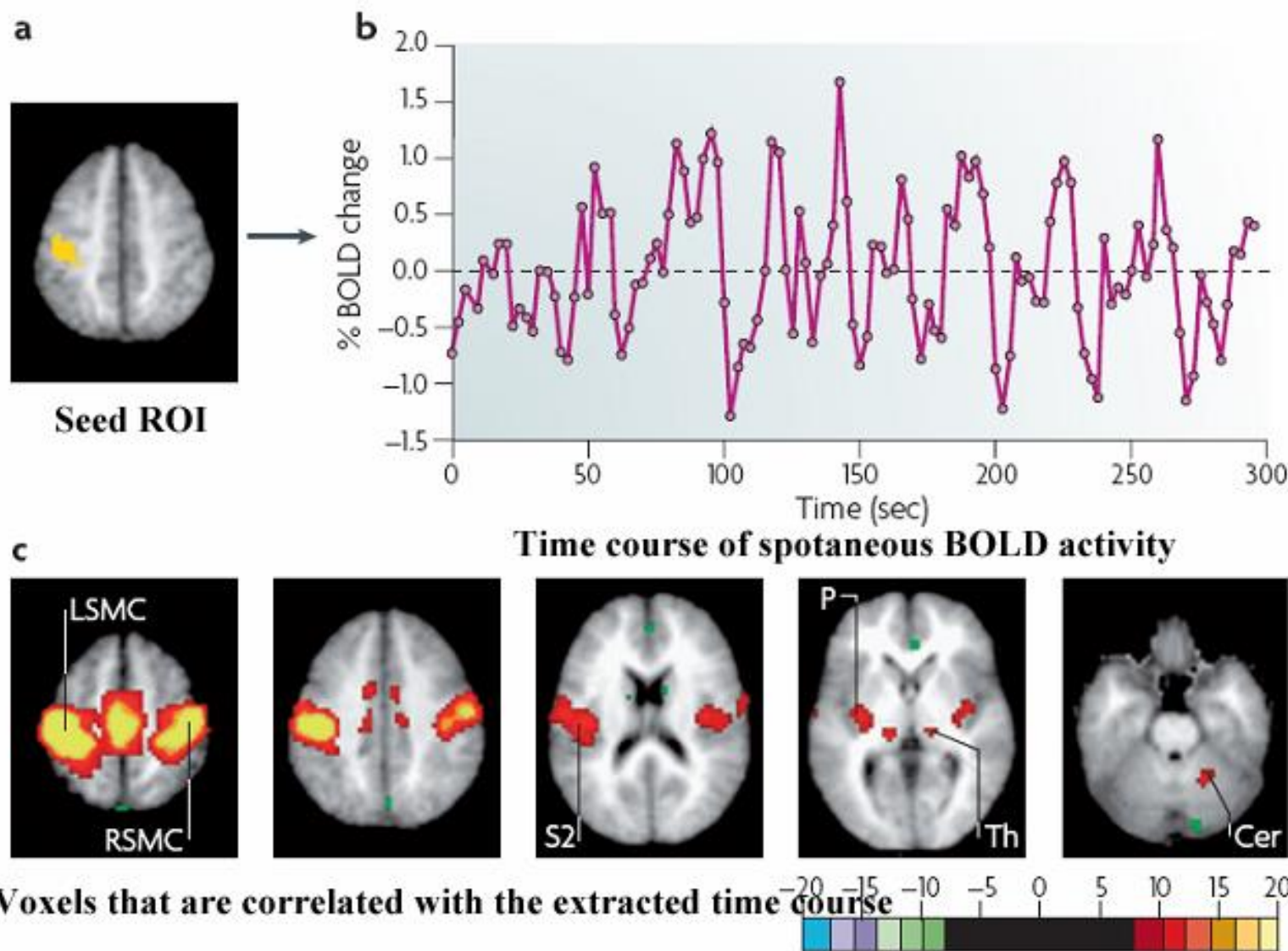
为什么要研究BOLD信号的自发性波动？

能量代谢：人脑占体重的2%，耗氧量为20%；任务诱导的代谢增加不足5%；绝大多数能量用于静息状态下的神经元活动

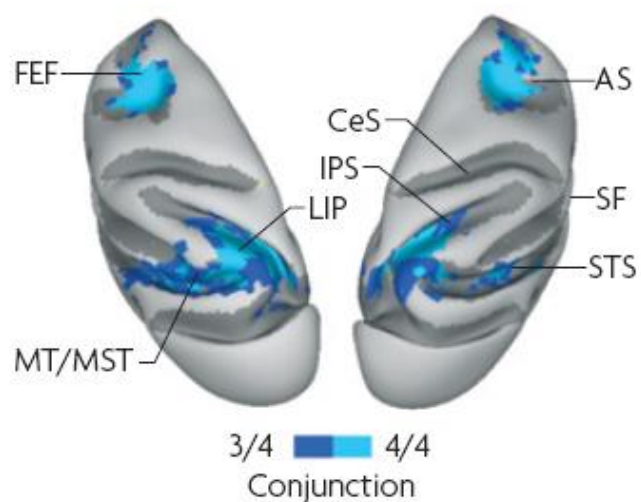
经验观察：在静息状态下，左侧与右侧感觉运动区的自发BOLD信号存在显著相关性，这种相关性也存在于运动前区及SMA

被试要求：闭上双眼，保持不动，不要睡觉，尽量不要思考具体问题

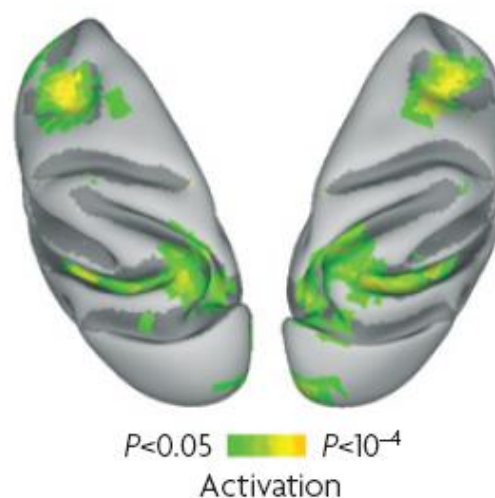
自发活动的同步性



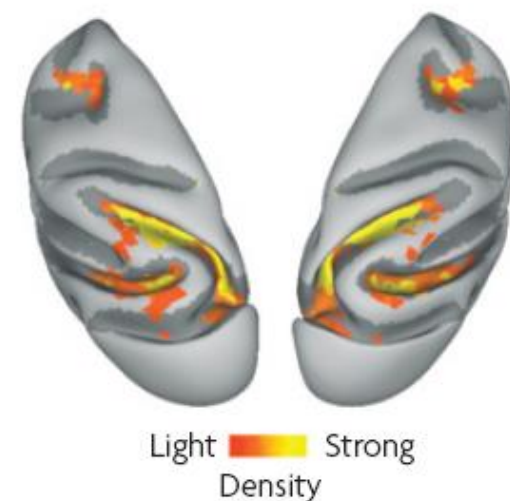
a Spontaneous-correlation pattern



b Evoked-response pattern

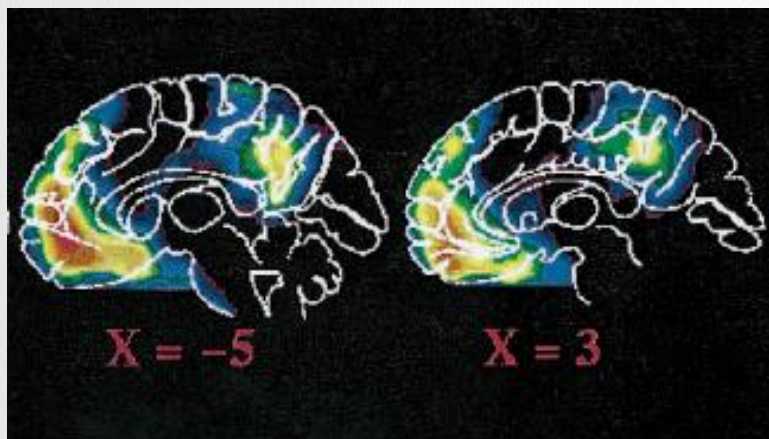


c Anatomical-connectivity pattern



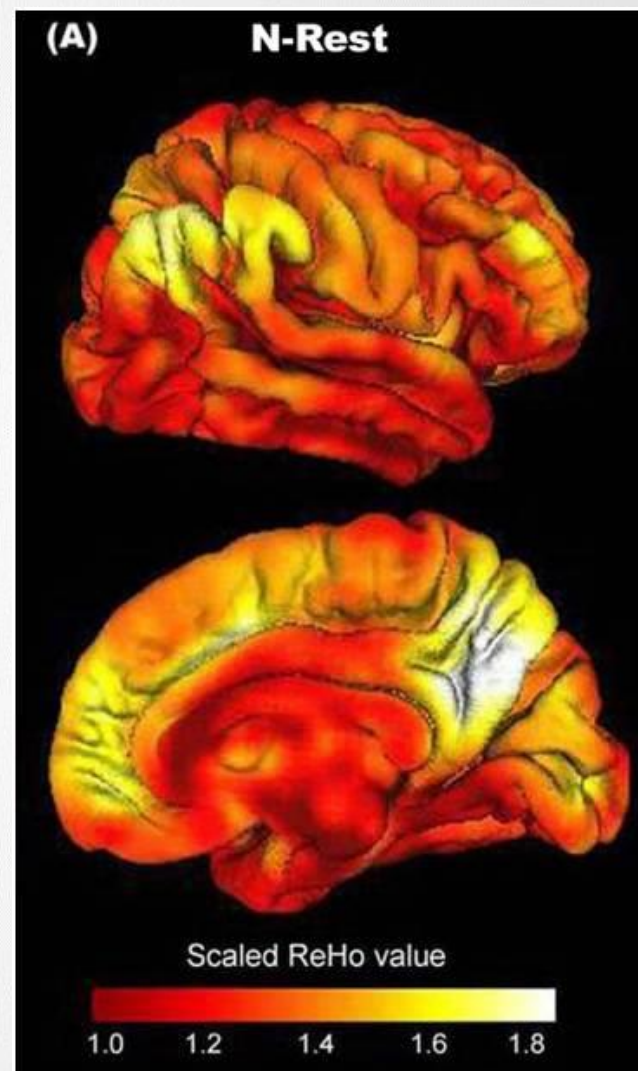
猴动眼系统

- 局部一致性 (ReHo)
- 脑活动以团块形式出现，而不是单个体素
- KCC评价一个体素与其相邻体素的时间序列的相似性
- 生理学意义：局部脑活动？
- 观察发现：默认网络内的ReHo值最高



Zang et al., 2004

Raichle et al., 2001



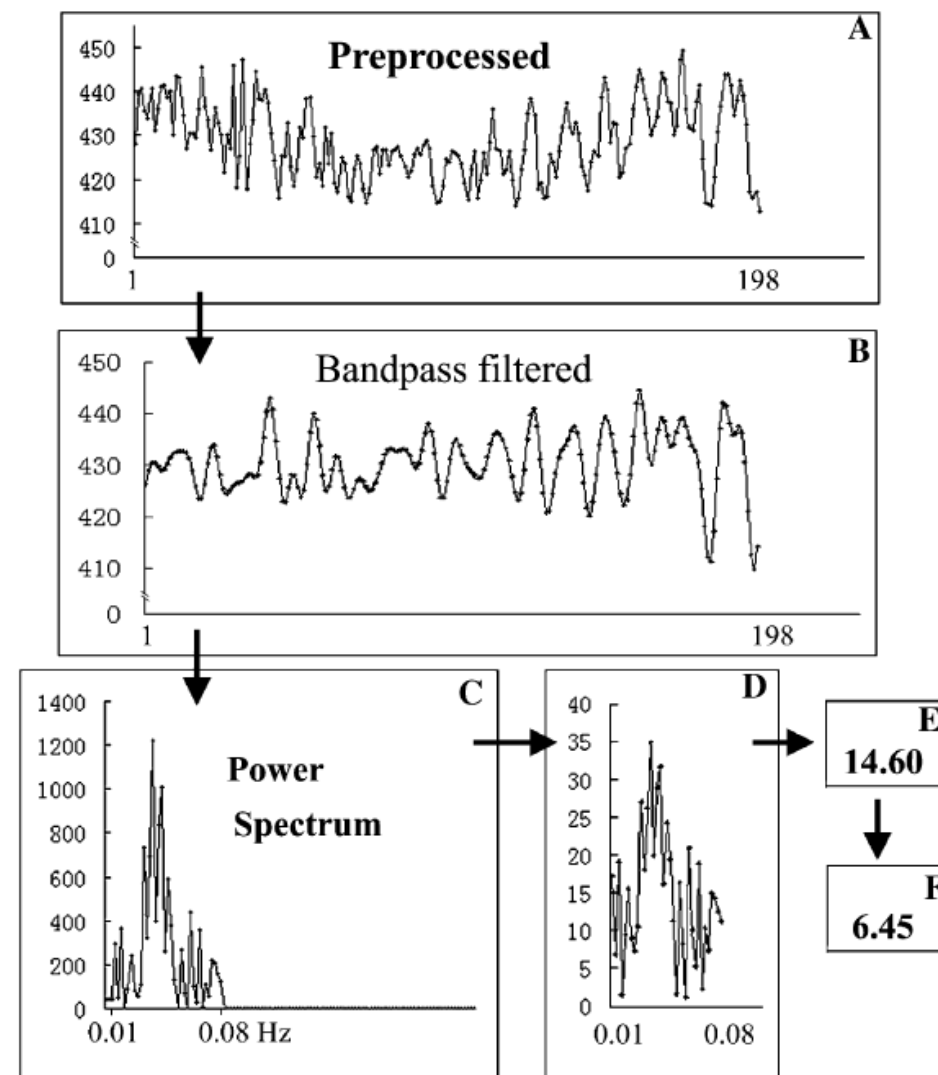
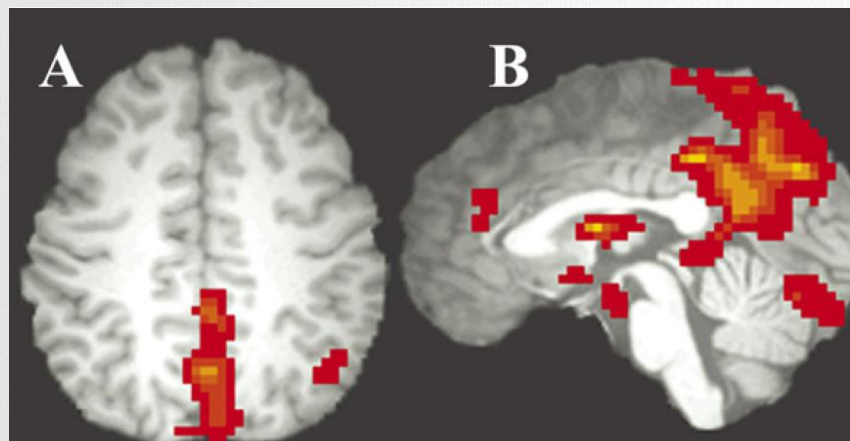
Wang et al., 2009

ALFF技术简介

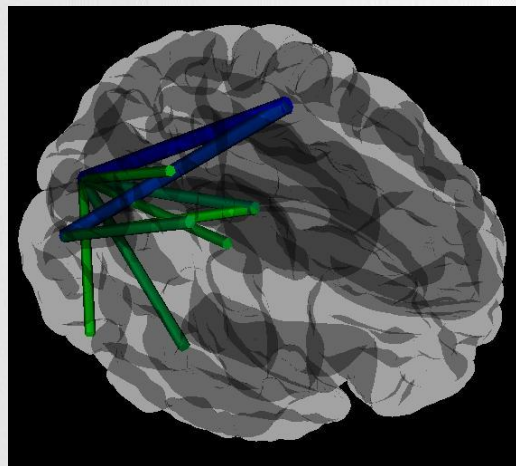


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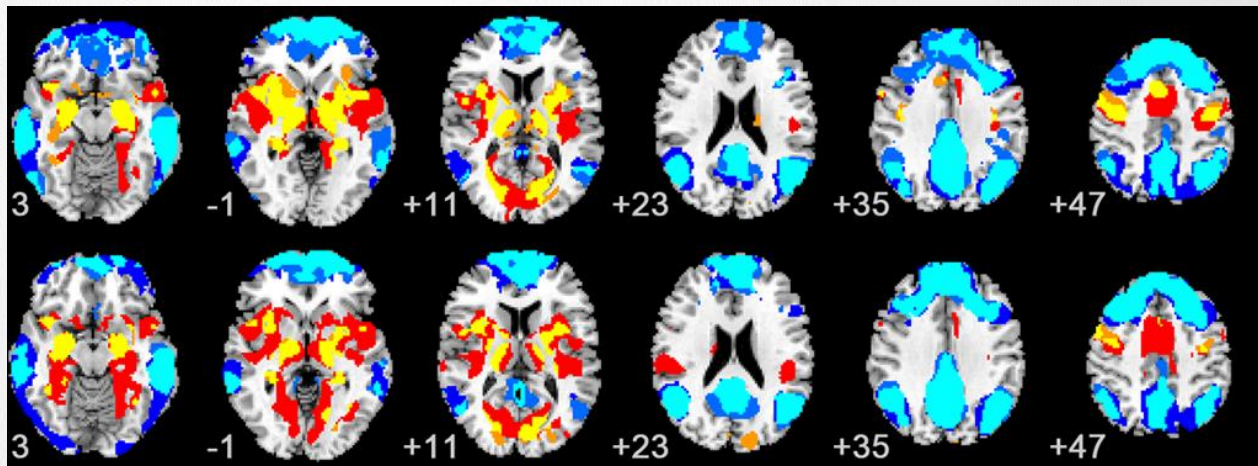
- 时间序列的信号强度
- 带通滤波 (0.01–0.08 Hz)
- 快速傅利叶变换的到功率谱
- ALFF: 功率谱的方根
- 0.01–0.08 Hz频段的平均ALFF
- 标准化的 ALFF



- 选定体素或ROI与其它体素或ROI时间序列的相关性
- 简单、敏感、易解释
- 必须定义种子区、不能同时研究多个系统、提取的波形并非独立变量



Yu et al., HBM 2008



Sang et al., Neuroimage 2012

● 优点

- 不需要定义种子区
- 获得统计上独立的组份
- 每个组分对应于一个空间图（噪声、神经网络）

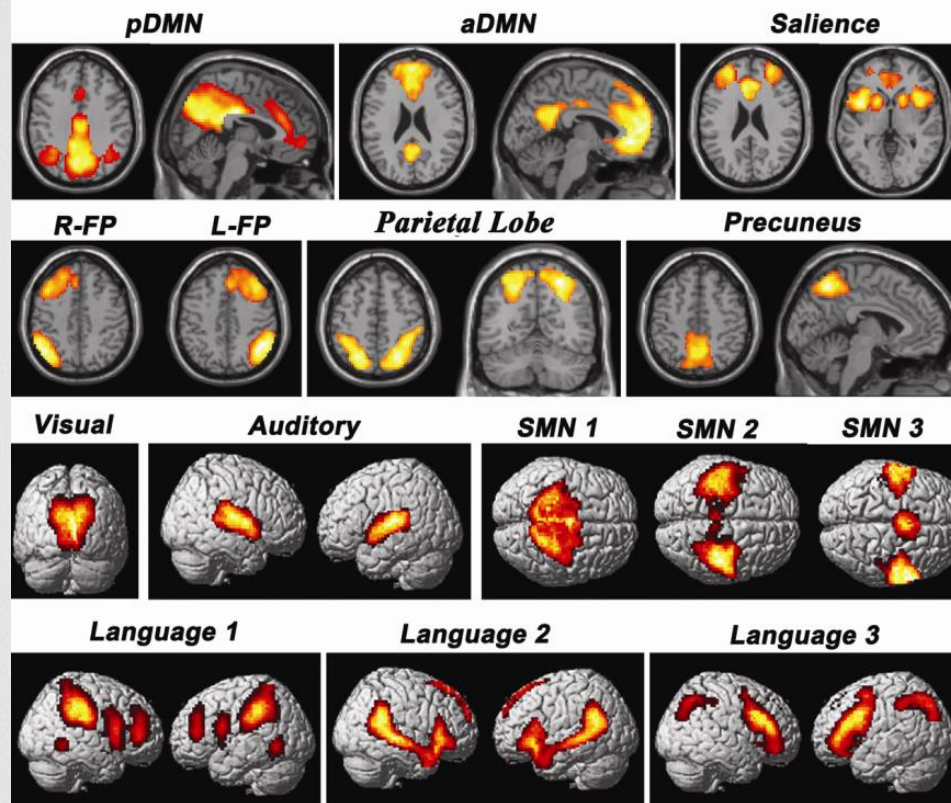
● 不足

- 结果与主观确定的组分数有关
- 组分的生物学意义需要根据人为定义的标准确定
- 解释复杂

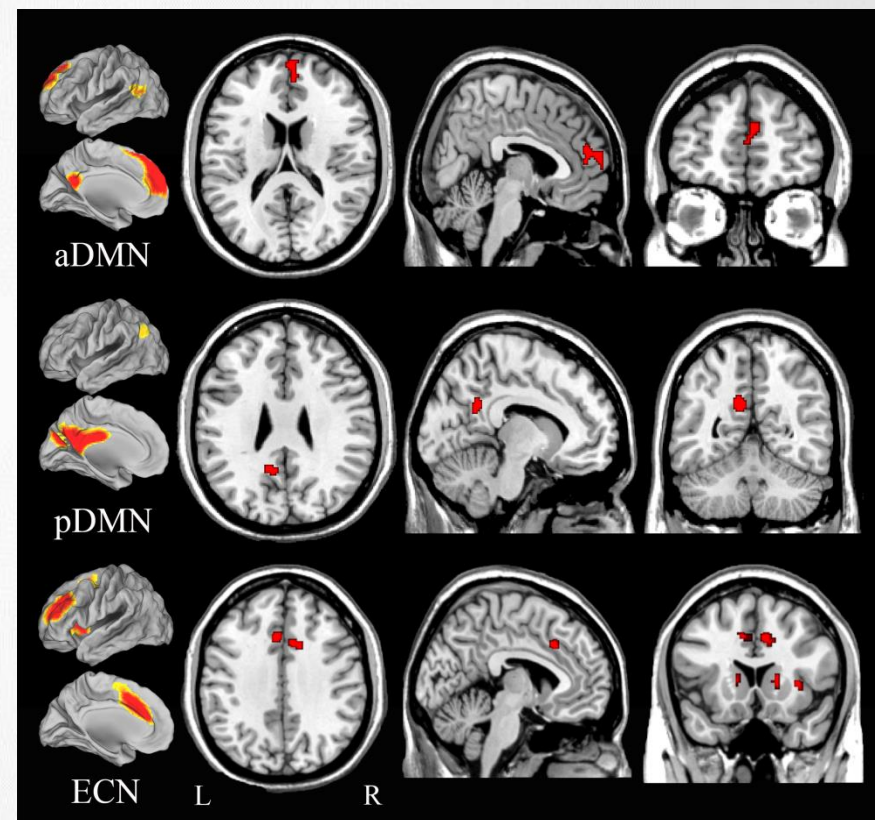
基于ICA的功能连接分析



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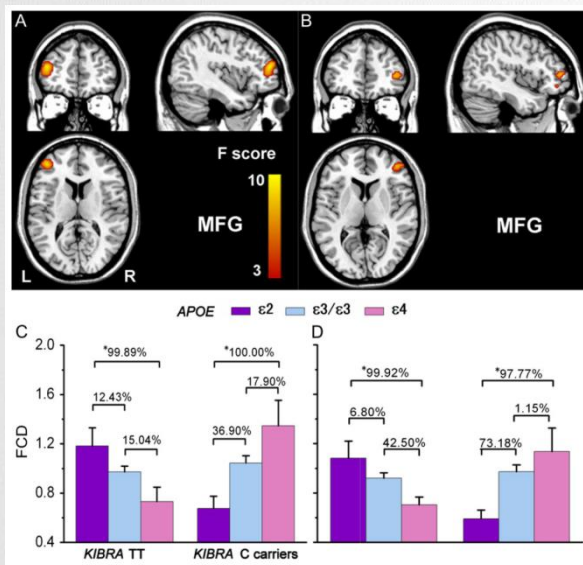


Xuan et al., Plos One 2012

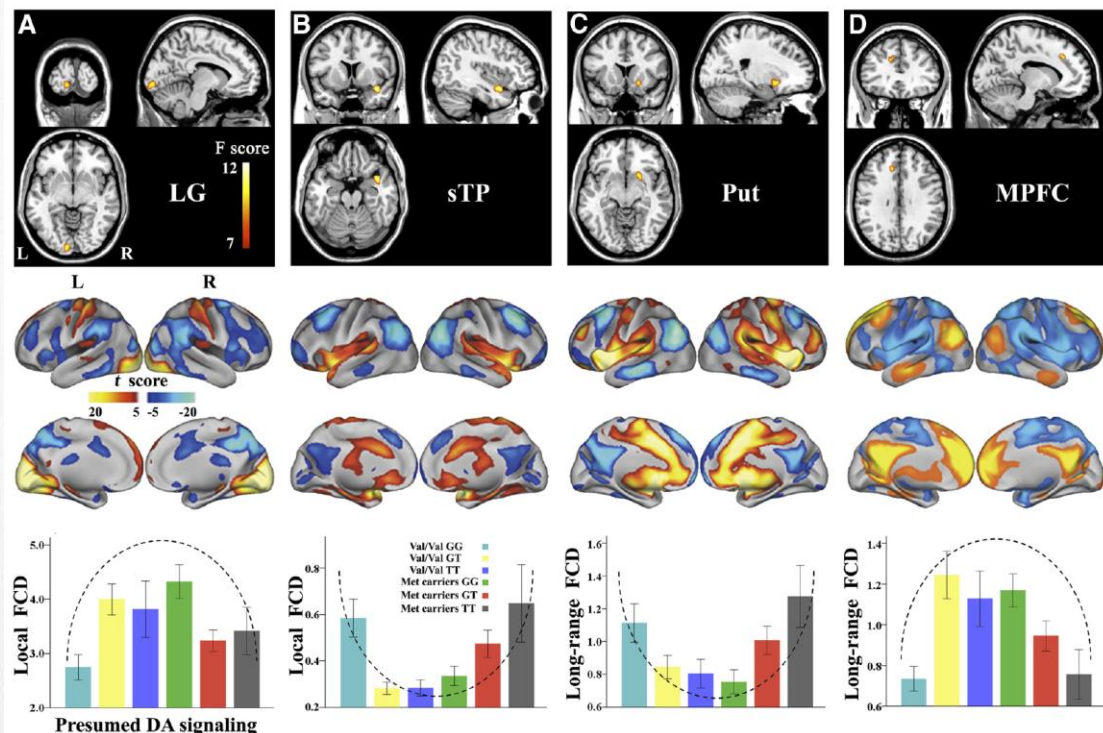


Wang et al., Neuroimage 2013

- 考察与选定体素存在功能连接的体素数目
- 需要人为选定一个功能连接阈值 (0.6)
- 长程FCD和短程FCD
- 功能连接强度 (FCS)

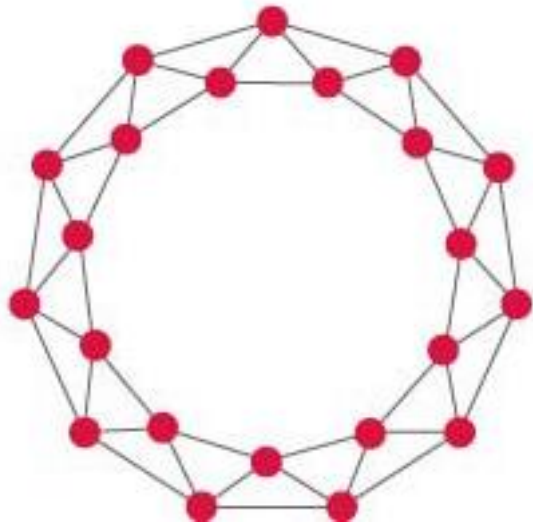


Zhang et al., Cereb Cortex 2016

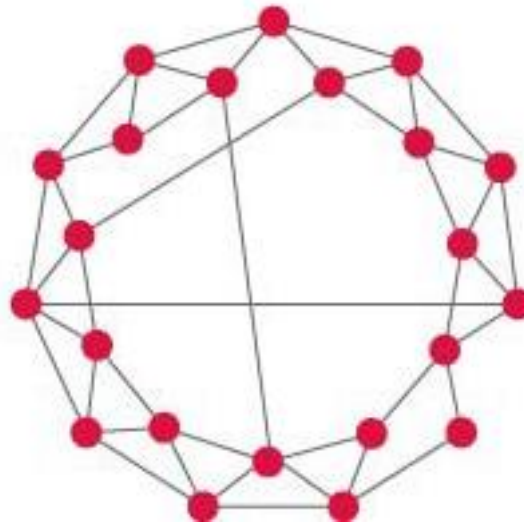


Tian et al., J Neuroscience 2013

REGULAR NETWORK



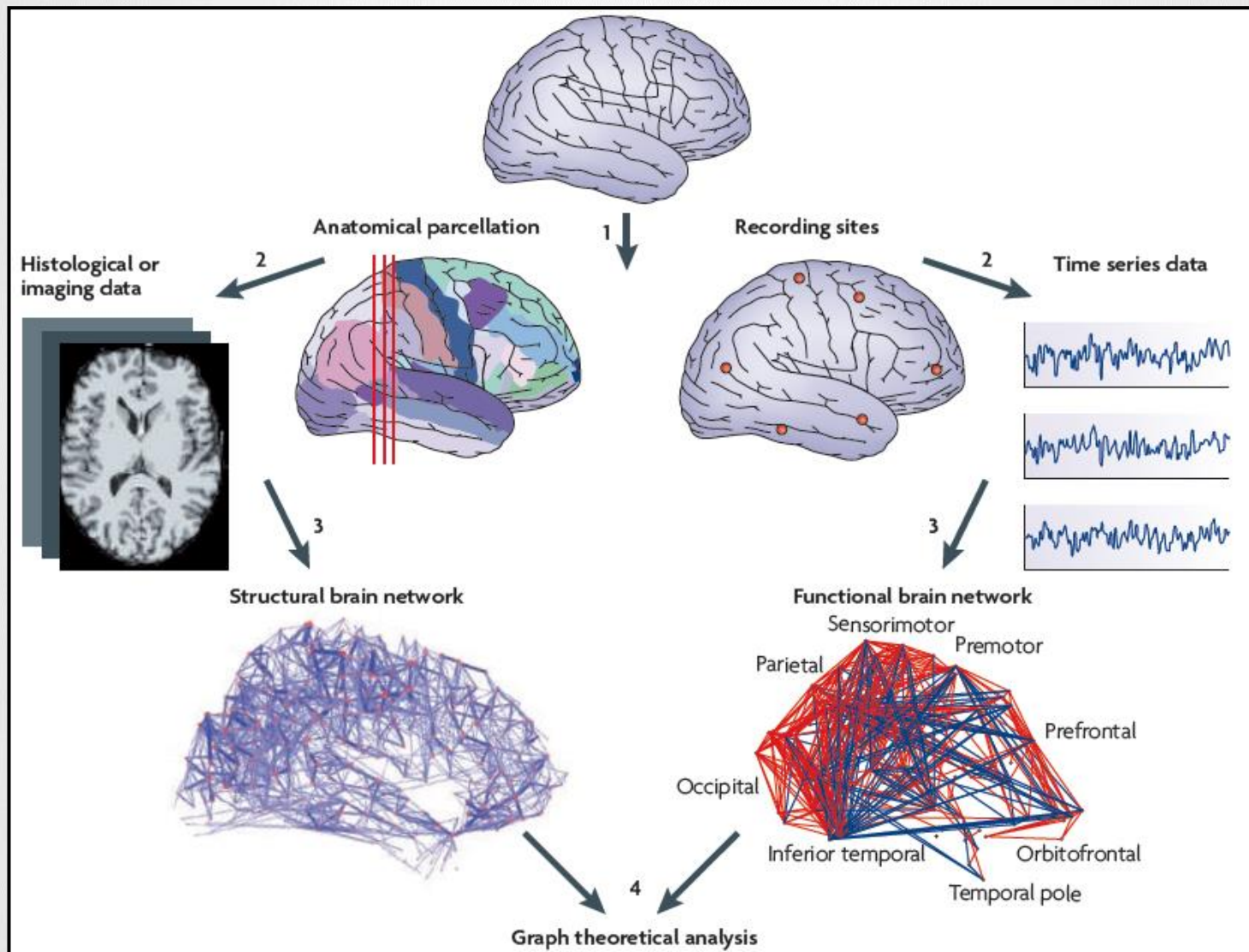
SMALL-WORLD NETWORK



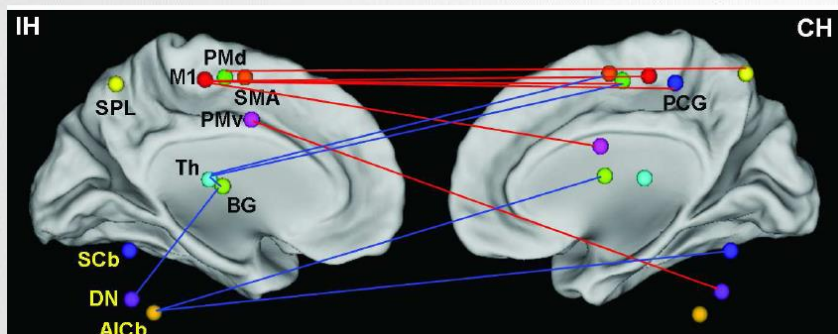
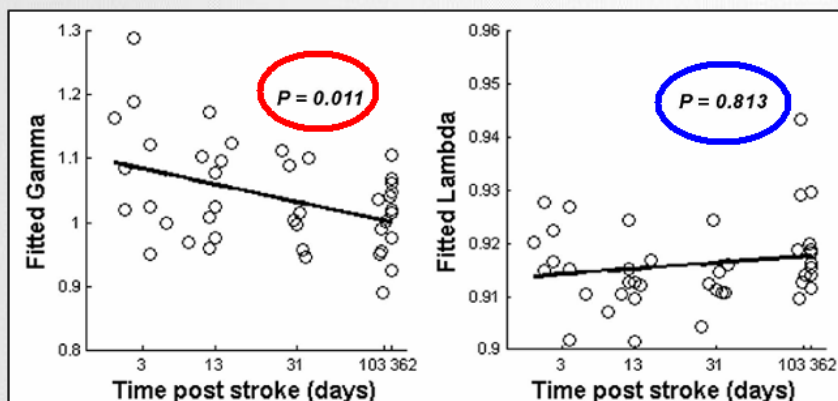
RANDOM NETWORK



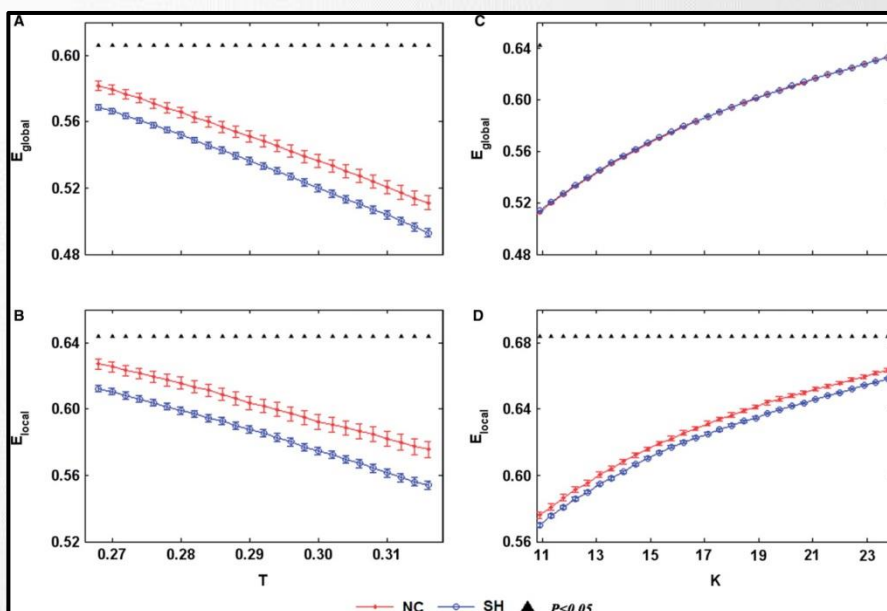
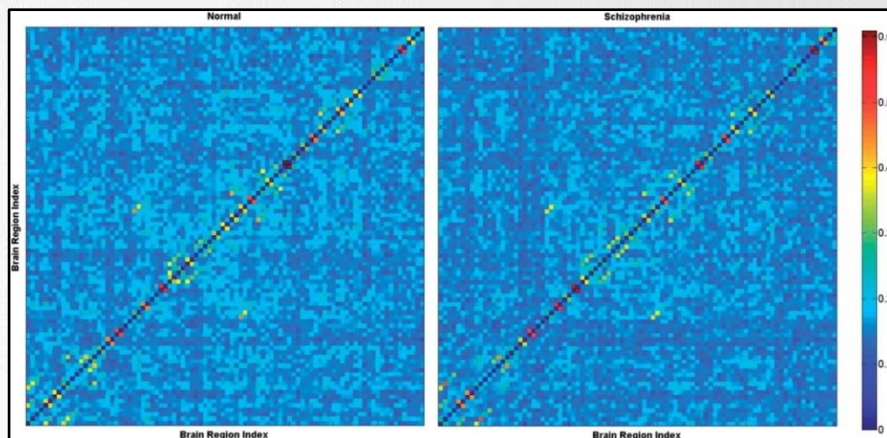
Small-world network



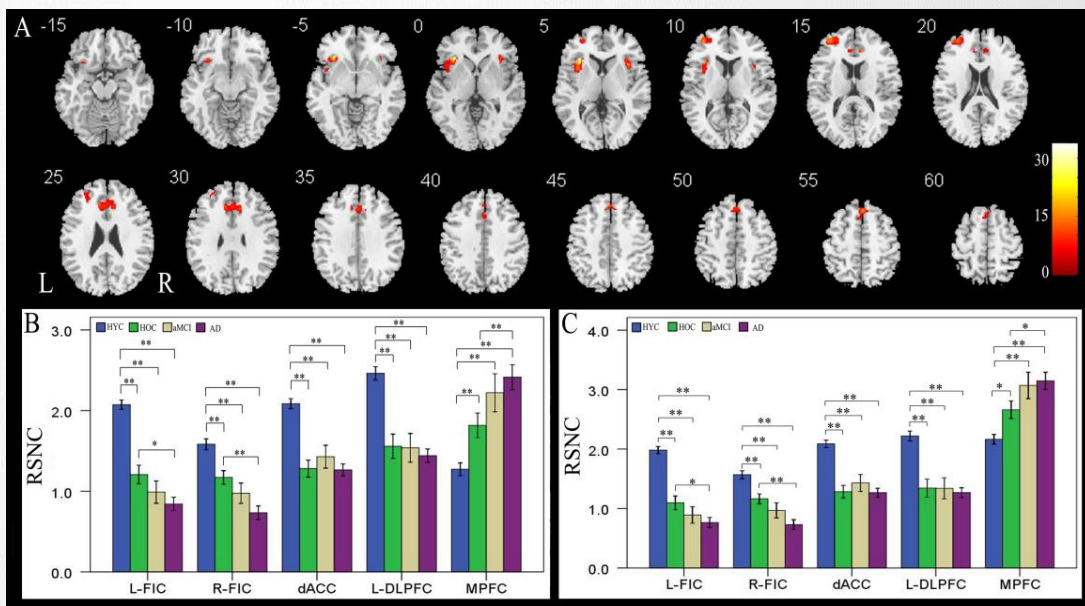
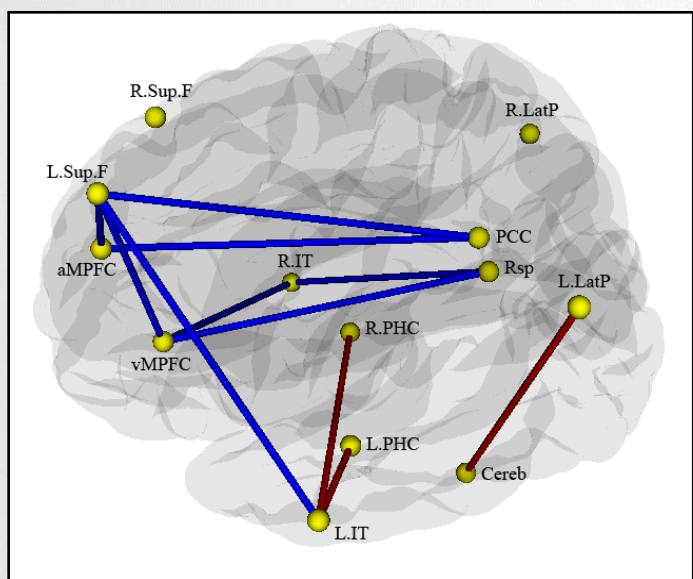
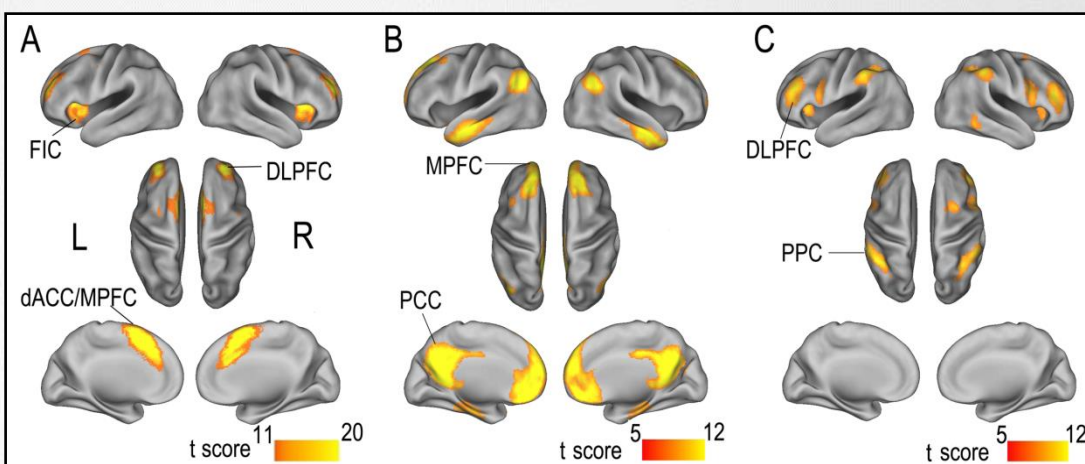
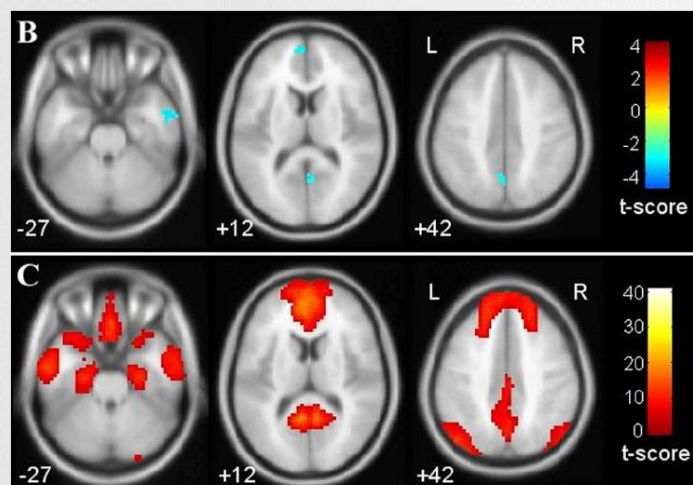
- ◆ 局部效率降低而全局效率不变
- ◆ 网络向随机化方向演变
- ◆ 左侧M1局部中心性增加



Wang et al., Brain 2010



Liu et al., Brain 2008

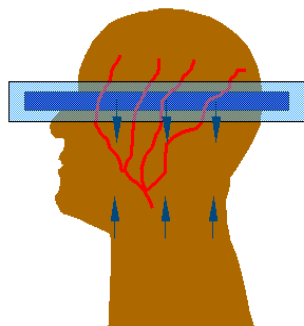


动脉自旋标记成像原理

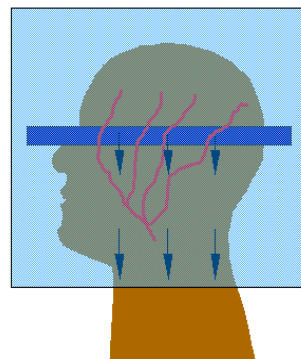


$$ASL_{\text{signal}} = \frac{(M_C - M_L)}{M_C}$$

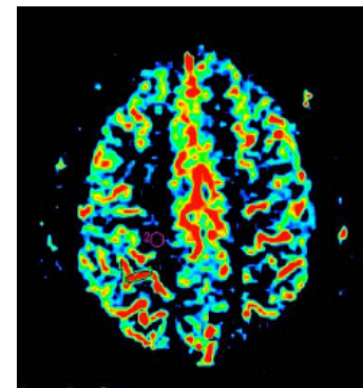
第一次采样
(标记组)



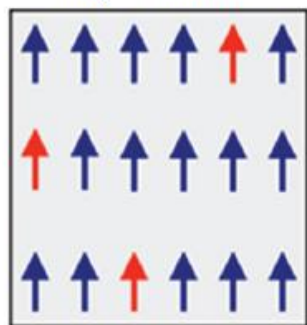
第二次采样
(对照组)



=

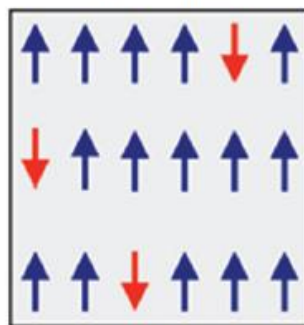


$M_C = \text{control}$



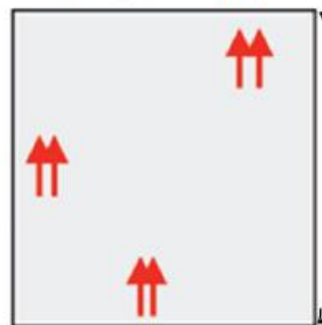
Signal ~ 18

$M_L = \text{label}$



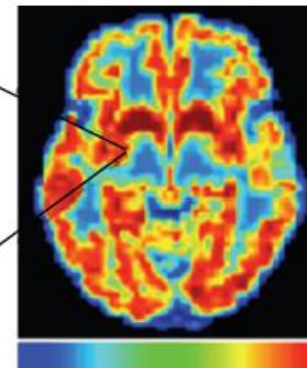
Signal ~ 12

$\Delta = (M_C - M_L)$



Signal ~ 6
Signal $\sim 2 \cdot \text{CBF}$

$\text{CBF} = f(T1, \lambda, \text{etc.}) \cdot \Delta / M_C$



mL/(100 g · min)

- **连续式标记 (CASL)**

优点：信噪比高、覆盖范围大

缺点：需要较长的RF，硬件要求高

- **脉冲式标记 (PASL)**

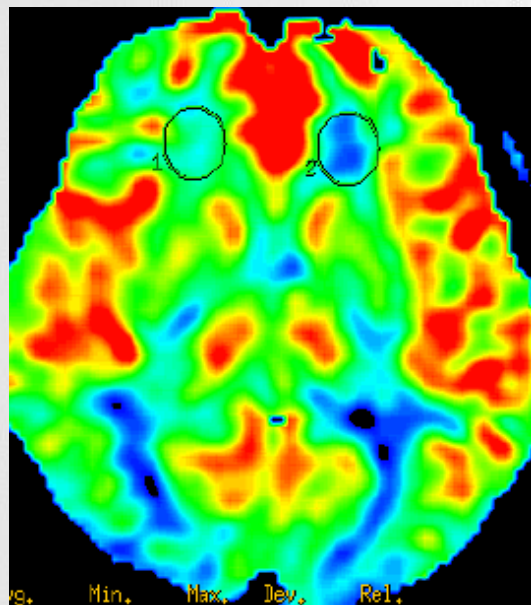
FAIR、EPISTAR

优点：易于实现、不受MT效应影响

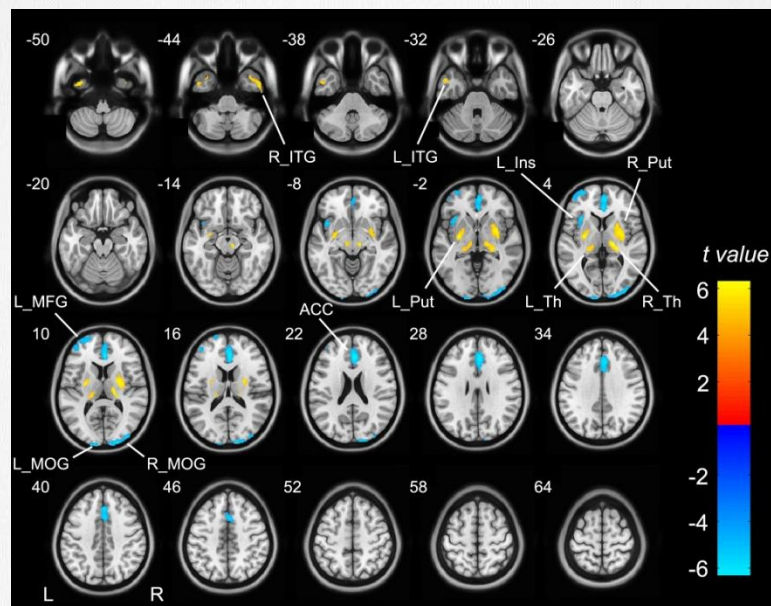
缺点：信噪比低、对通过时间敏感、层间伪影大

- **伪连续式标记 (pCASL)**

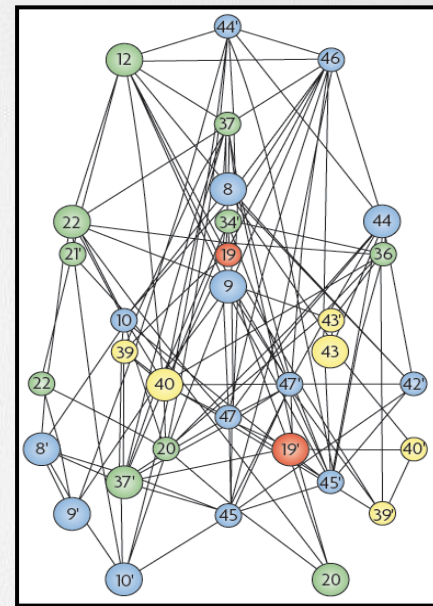
标记效率高、覆盖范围大、硬件要求适中



基于ROI分析



基于体素分析



网络分析



天津医科大学

TIANJIN MEDICAL UNIVERSITY

谢谢聆听!